Networking - Devices and tools

📡 Routers, Switches, Hubs

🔹 1. Hub

📌 What it is

- A **basic networking device** that connects multiple devices in a LAN.
- Operates at the **Physical Layer (Layer 1) of OSI**.

📌 How it Works

- When a device sends data, the hub **broadcasts it to all connected devices**.
- Only the intended recipient accepts it; others discard it.

📌 Characteristics

- No intelligence → cannot check MAC or IP.
- Creates a **single collision domain** (data collisions likely).
- Cheap but inefficient.

📌 Use Cases

- Rarely used today.
- Sometimes in very small networks or for testing.

🔹 2. Switch

📌 What it is

- A **smarter device than a hub** that connects devices in a LAN.
- Operates at the **Data Link Layer (Layer 2)**.
- Some advanced switches can also operate at **Layer 3 (routing features)**.

📌 How it Works

- Maintains a **MAC address table**.
- Forwards frames only to the specific port/device that needs it.

📌 Characteristics

- Reduces collisions (each port = separate collision domain).
- Supports **full-duplex** communication (send & receive simultaneously).
- Faster and more secure than hubs.

📌 Use Cases

- Common in LANs (homes, offices, data centers).
- Backbone of most modern networks.

🔹 3. Router

📌 What it is

- A device that **connects multiple networks** (e.g., LAN to WAN/Internet).
- Operates at the **Network Layer (Layer 3)**.

📌 How it Works

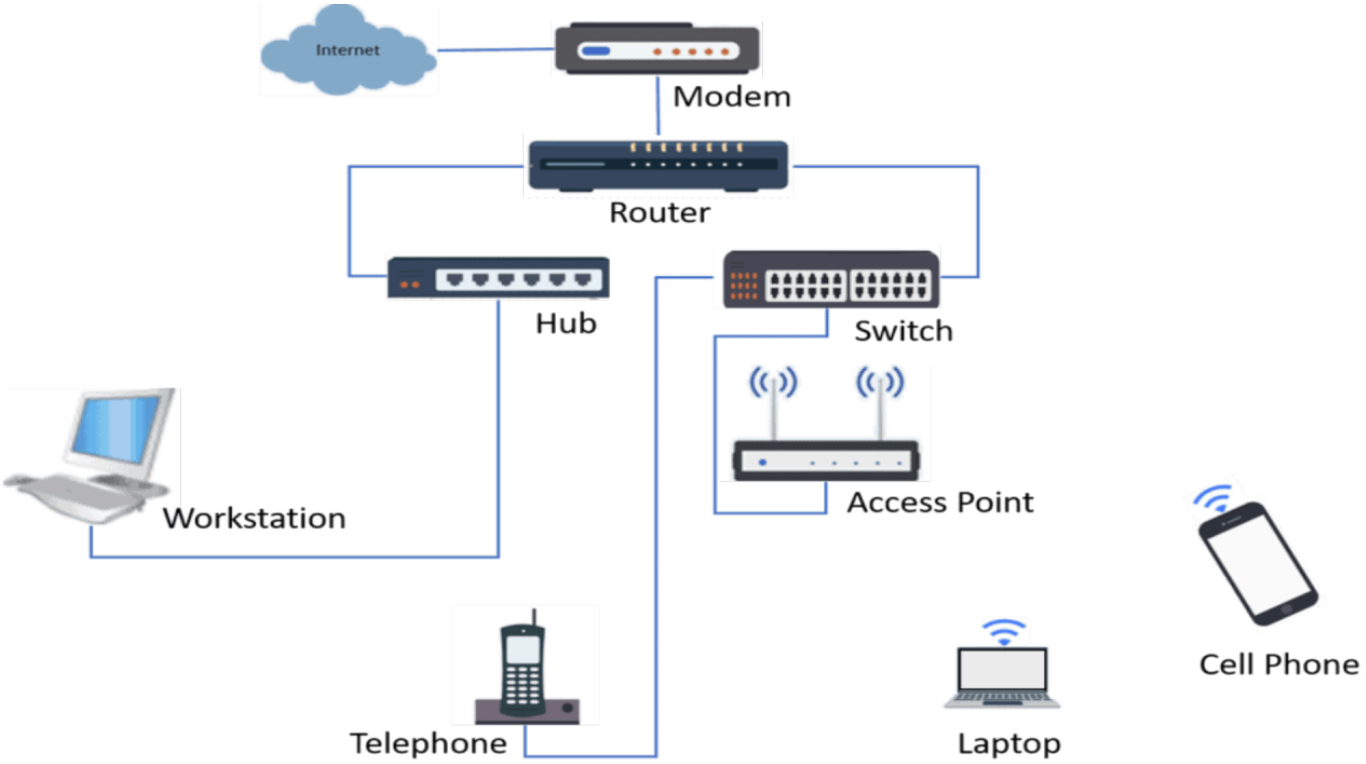
- Uses **IP addresses** to forward packets.
- Determines the **best path** for data using routing tables and protocols.

📌 Characteristics

- Separates **broadcast domains**.
- Can provide **NAT (Network Address Translation)** and **DHCP**.
- Supports firewall and VPN features in modern routers.

📌 Use Cases

- Home routers (connects your devices to the Internet).
- Enterprise routers (connect branches, manage large networks).



Firewalls, IDS, IPS

🔹 1. Firewall

📌 What it is

- A **security device/software** that monitors and controls network traffic.
- Acts as a **barrier** between trusted (internal) and untrusted (external) networks.
- Works at multiple OSI layers depending on type.

📌 Types of Firewalls

1. **Packet-Filtering Firewall**
   - Works at **Network Layer (Layer 3)**.
   - Allows/blocks traffic based on **IP addresses, ports, protocol**.
2. **Stateful Firewall**
   - Works at **Network + Transport Layers (Layers 3 & 4)**.
   - Keeps track of connection states (established, new, related).
3. **Application Firewall (Proxy Firewall)**
   - Works at **Application Layer (Layer 7)**.
   - Inspects actual application data (e.g., HTTP requests).
4. **Next-Generation Firewall (NGFW)**
   - Combines traditional firewall + IDS/IPS + deep packet inspection.

📌 Use Cases

- Protect internal networks.
- Block malicious or unauthorized traffic.
- Enforce security policies.

🔹 2. IDS (Intrusion Detection System)

📌 What it is

- A **monitoring system** that detects malicious activity or policy violations.
- Works **passively** → only detects and alerts, does not block.

📌 Types of IDS

1. **NIDS (Network-based IDS)** → Monitors traffic across entire network.
2. **HIDS (Host-based IDS)** → Monitors activity on a single device.

📌 How it Works

- Compares traffic against a **signature database** (known attack patterns).
- Uses **anomaly detection** (flags unusual behavior).

📌 Limitation

- Cannot block attacks (alert-only).

🔹 3. IPS (Intrusion Prevention System)

📌 What it is

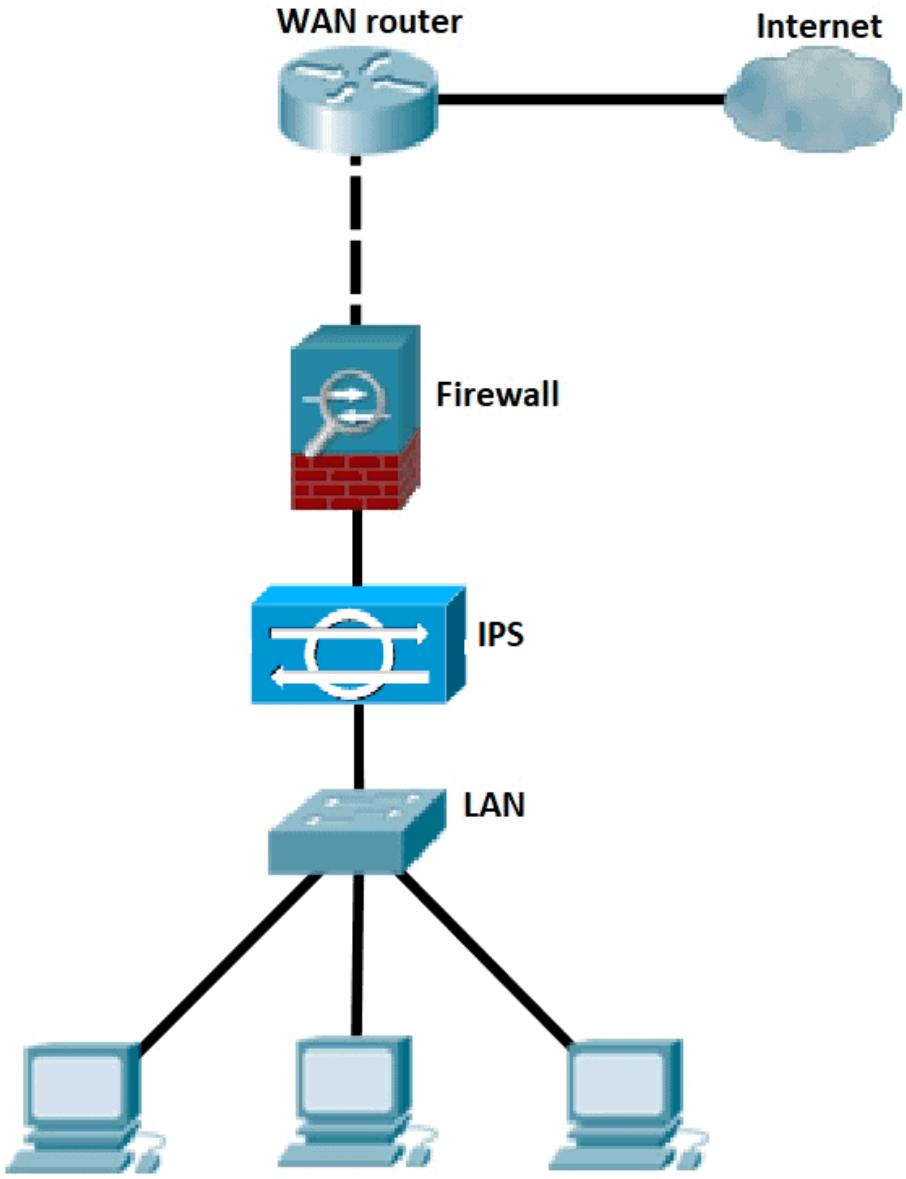
- Similar to IDS, but **active** → detects and blocks malicious traffic.
- Placed **inline** (between sender and receiver).

📌 How it Works

- Detects malicious traffic in real time.
- Drops or modifies packets that match attack patterns.

📌 Types

- **NIPS** → Network-based IPS.
- **HIPS** → Host-based IPS.



Load Balancers

📌 What is a Load Balancer?

- A **network device or software** that distributes incoming traffic across multiple servers.
- Ensures no single server is overwhelmed, improving **availability, performance, and reliability**.
- Works at **Layer 4 (Transport)** or **Layer 7 (Application)**.

🔹 Why Load Balancers are Needed

- Without load balancing → one server could get overloaded while others sit idle.
- With load balancing → traffic is shared, users get faster response, and services remain available even if one server fails.

🔹 Types of Load Balancers

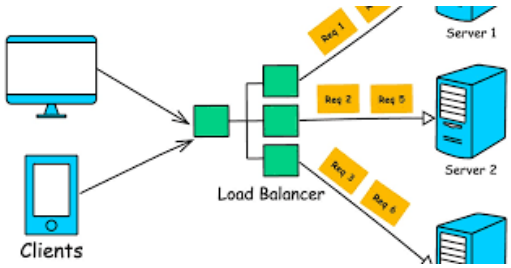
1. **Layer 4 Load Balancer**
   - Works at the **Transport Layer**.
   - Balances traffic based on **IP address & port**.
   - Faster but less flexible.
2. **Layer 7 Load Balancer**
   - Works at the **Application Layer**.
   - Can inspect actual content (e.g., HTTP headers, cookies, URLs).
   - Can route requests intelligently (e.g., send image requests to a media server).

🔹 Load Balancing Algorithms

1. **Round Robin**
   - Requests sent to servers in a circular order.
2. **Least Connections**
   - Sends traffic to the server with the fewest active connections.
3. **IP Hash**
   - Chooses server based on client's IP address (same client always goes to same server).
4. **Weighted Round Robin**
   - Servers with more capacity get more requests.

🔹 Benefits of Load Balancers

- **Scalability** → add more servers to handle more users.
- **High Availability** → if one server goes down, others continue serving.
- **Improved Performance** → prevents overload, reduces response time.
- **Security** → can hide internal servers, mitigate DoS attacks.



Proxies (Forward & Reverse)

📌 What is a Proxy?

- A **proxy server** acts as an **intermediary** between a client and the destination server.
- The client sends requests to the proxy → the proxy forwards them to the destination → returns the response back to the client.
- Works at the **Application Layer (Layer 7)**.

🔹 1. Forward Proxy

📌 How it Works

- Sits **in front of the client**.
- The destination server sees requests as if they are coming from the proxy, not the client.

📌 Use Cases

- **Privacy/Anonymity** → hides client's IP.
- **Access Control** → restricts client access to certain websites.
- **Caching** → stores frequently requested content to speed up responses.

Example:

- User in office → request goes to corporate proxy → proxy forwards to Internet.

🔹 2. Reverse Proxy

📌 How it Works

- Sits **in front of the server(s)**.
- Clients don't directly connect to backend servers, but to the reverse proxy.

📌 Use Cases

- **Load Balancing** → distribute traffic to multiple servers.
- **Security** → hides backend server details from clients.
- **Caching/Compression** → speeds up responses.
- **SSL Termination** → handles encryption/decryption before passing traffic to servers.

Example:

- Client requests `www.example.com` → reaches reverse proxy → proxy forwards to one of many backend servers.

VPNs & Tunneling

🔹 What is a VPN?

- **VPN (Virtual Private Network)** creates a **secure, encrypted connection** between a device and a remote network over the Internet.
- It hides the real IP address and encrypts traffic → providing **privacy, security, and anonymity**.

🔹 How VPN Works

1. Device connects to a **VPN server**.
2. All traffic is **encrypted** and sent through a secure tunnel.
3. VPN server forwards the request to the destination (e.g., website).
4. The response comes back through the same tunnel → decrypted at the device.

🔹 VPN Protocols

Protocol	Layer	Features
PPTP (Point-to-Point Tunneling Protocol)	Layer 2	Old, fast but insecure
L2TP/IPsec (Layer 2 Tunneling Protocol + IPsec)	Layer 2 + 3	Stronger security, widely used
OpenVPN	Layer 4 (TCP/UDP)	Open-source, secure, flexible
IKEv2/IPsec	Layer 3	Fast, stable, good for mobile
WireGuard	Layer 3	Modern, lightweight, very fast & secure

🔹 What is Tunneling?

- Tunneling is the **process of encapsulating one protocol inside another** to securely transmit data.
- VPN uses tunneling to wrap original traffic in an encrypted “envelope”.

Types of Tunneling:

1. **Voluntary Tunneling** → Initiated by the user (via VPN client).
2. **Compulsory Tunneling** → Initiated by the ISP or network provider.

🔹 Benefits of VPNs

- **Security** → Encrypts traffic, protects against sniffing.
- **Privacy** → Hides browsing activity and real IP.
- **Remote Access** → Securely connect to corporate networks.
- **Bypass Restrictions** → Access geo-blocked or censored sites.

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extra essentials

🔹 1. VLANs (Virtual LANs)

📌 What is a VLAN?

- **VLAN (Virtual LAN)** is a logical segmentation of a physical LAN into smaller, isolated networks.
- Configured on **switches (Layer 2)**.

📌 Why VLANs?

- Improves **security** (separate traffic, e.g., HR vs Guest).
- Reduces **broadcast traffic**.
- Adds **flexibility** (devices can be grouped logically, not physically).

📌 Example

- One office switch, but:
  - VLAN 10 → HR department.
  - VLAN 20 → IT department.
  - VLAN 30 → Guest Wi-Fi.

📌 Security Context

- Prevents unauthorized access between departments.
- Misconfiguration can lead to **VLAN hopping attacks**.

🔹 2. Wireless Networking & Security Basics

📌 Wi-Fi Standards (802.11 family)

- **802.11a/b/g** → older, slower.
- **802.11n (Wi-Fi 4)** → faster, supports MIMO.
- **802.11ac (Wi-Fi 5)** → high-speed, widely used.
- **802.11ax (Wi-Fi 6)** → latest, supports higher efficiency & capacity.

📌 Wi-Fi Security Protocols

1. **WEP (Wired Equivalent Privacy)**
   - First Wi-Fi security standard.
   - Very weak, easily cracked with tools like Aircrack-ng.
2. **WPA (Wi-Fi Protected Access)**
   - Temporary improvement over WEP, but still weak.
3. **WPA2**
   - Stronger encryption (AES).
   - Still vulnerable to attacks (KRACK).

4. **WPA3**
   - Latest and most secure.
   - Uses SAE (Simultaneous Authentication of Equals).

📌 Security Context

- Weak Wi-Fi = easy entry point for attackers.
- Rogue APs, Evil Twin attacks, and Wi-Fi sniffing are common.